AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

TM298
Operating Systems

Final Examination
Fall Semester 2025/2026

Date: 20/Jan./2026

umber of Exam Pages: 3 cluding this cover sheet)	Time Allowed: 2 Hours

nstructions:

- Total Marks:**50**
- This exam consists of **three** parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Programmable Calculators** are **not** allowed.

CamScanner

Part 1: Multiple Choice Questions: [10 marks]

Question 1: Choose the most correct answer for each of the following:

1. _____ presents a uniform device-access interface to the I/O subsystem.
   a) An I/O system
   b) A bus
   c) A device driver
   d) A device
2. Which of the following is not a type of system call used for Protection?
   a) Get Permissions
   b) Control Access
   c) Set permissions
   d) Set system data.
3. The correct operations order for protecting a critical section using **Mutex** locks is:
   a) signal() followed by wait()
   b) release() followed by acquire()
   c) wait() followed by signal()
   d) acquire() followed by release()
4. The priority of a process _____ if the scheduler assigns it a static priority.
   a) Will change.
   b) May change.
   c) Remain unchanged.
   d) Any of the choices.
5. CPU scheduling is the basis of ____.
   a) Multiprocessing OS
   b) Multiprogramming OS
   c) Large memory sized OS
   d) None of the mentioned
6. In Priority Scheduling, a higher numbered priority typically indicates:
   a) Higher priority
   b) No specific priority level
   c) Lower priority
   d) Both choices, b and c.
7. Which scheduling algorithm is best suited for time-sharing systems?
   a) Round-Robin
   b) First-Come, First-Served
   c) Shortest Job Next
   d) Priority Scheduling.
8. If a system is in the unsafe mode, this means that we have:
   a) Deadlock
   b) A possibility of a deadlock
   c) No deadlock
   d) Either a or b
9. Which of the following is an example of an open-source operating system?
   a) Windows 11
   b) Mac OS
   c) Linux
   d) Google Drive.
10. In Unix, which system call creates the new process?
    a) create
    b) new
    c) fork
    d) Any of the choices

Part 2: Short Essay questions: [15 marks]

Question 2: [3 marks]

What is the role of the memory management unit (MMU) in an operating system?

Question 3: [4 marks]

a. Explain the concept of resource allocation graph (RAG). (2 marks)
b. What do the nodes and directed edges represent? (2 marks)

Question 4: [4 marks]

Write the two methods that implement the simple **semaphore** wait() and signa operations on global variable **S**.

CamScanner

Question 5: [4 marks]

a- What is the atomic variable? (2 marks)

b- Define the two types of semaphore, Counting and Binary. (2 marks)

Part 3: Essay/Problem Solving questions: [25 marks]

Question 6: [6 marks]

Explain the difference between global replacement and local replacement in virtual memory management and list one advantage for each.

Question 7: [6 marks]

Calculate the **Effective Access Time (EAT)** in **nanoseconds** and **milliseconds** for a demand paging system, where:

- Memory access time = **100 nanoseconds**
- Average page-fault service time = **10 milliseconds**
- Page-fault rate (**p**) = **0.6 / 100**

(Show the full steps of your calculations)

Question 8: [6 marks]

Consider the following segment table:

Segment	Base	Limit
0	100	200
1	400	110
2	50	125
3	600	180
4	790	190

Find the physical addresses for the following logical addresses and deduce whether this address will be of legal reference or not. (Show the calculations)

a) Segment: 1, Logical address: 100 (2 marks)

b) Segment: 2, Logical address: 135 (2 marks)

c) Segment: 4, Logical address: 190 (2 marks)

Question 9: [7 marks]

Consider the following page reference string: 7,5,6,3,7,3,6,5,2,8,6,4

Ref String	7	5	6	3	7	3	6	5	2	8	6	4
f1												
f2												
f3												
H/PF												

Assume that there are **three frames**, and the **Optimal algorithm** was used. Answer the following questions:

a- Fill in the table above after copying it to your answer booklet. (6 marks)

b- Find the number of Page Faults (PF). (1 mark)

Instructions:

- in the last row, you should indicate whether we have a hit (**H**) or a page fault (**PF**)
- In the last column(s), if you do not have future pages, use **FIFO**.

End of questions

CamScanner